ARAB YOUTH HACKATHON 2026

EARTH OBSERVATION | GEOAI | SUSTAINABLE CITIES

UrbanTwin AI

Earth Observation Intelligence for Climate-Resilient Arab Cities

FROM SATELLITE OBSERVATIONS
TO URBAN PLANNING INTELLIGENCE



TEAM
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THEME
Urban Expansion, Land Use Change & Heat Risk



COUNTRY
Jordan

THE PROBLEM

Rapid Urban Expansion in Arab Cities

More built-up areas

Less green spaces

Higher surface temperatures

Changing land use



A GROWING CHALLENGE

Fast-changing cities require spatial insights to guide smarter, more resilient planning.

WHO IS AFFECTED?

- Municipalities & planners
- Urban designers & developers
- Environment & climate experts
- Local communities
- Planning consultancies
- Researchers & sustainability organizations

BUSINESS USE CASE & END USER

BUSINESS & TARGET USERS

Target Customers

Municipalities • Urban Planning Authorities • Smart-City Programs
Urban Developers • Engineering & Planning Consultancies
Climate & Sustainability Organizations • Research Institutions

Business Model

Urban Intelligence as a Service

- Recurring urban monitoring subscriptions
- Automated reports and alerts
- Interactive intelligence dashboards
- Custom Digital Twin deployments
- Data and analytics APIs
- Enterprise analytics and integrations

The product can begin with a selected urban area and scale toward **multi-city Urban Intelligence across the Arab region.**



END USERS

- Municipal planning teams
- Urban designers and planners
- Planning consultancies and developers
- Climate-resilience and sustainability teams

EXPECTED IMPACT

EXPECTED IMPACT

UrbanTwin AI will enable stakeholders to turn satellite data into practical insights, supporting more informed, resilient and sustainable urban planning decisions.



01

MONITOR

Urban expansion and land-use change over time.

02

IDENTIFY

Areas experiencing increasing heat exposure, development pressure or green-space loss.

03

PRIORITIZE

Locations requiring further investigation, planning attention or potential intervention.

04

SUPPORT

Evidence-based urban planning and climate-resilience decisions.

05

CONNECT

Earth Observation intelligence with practical urban planning workflows.

The platform is designed to provide **consistent, spatially explicit and repeatable urban intelligence,** helping bridge the gap between Earth Observation data and practical planning decisions.

05 • DATA & SOURCES

SATELLITE DATA FOR URBAN INSIGHTS

The current PoC combines optical surface information with a satellite-derived thermal reference for a UAE study area.

SENTINEL-2C · CURRENT PoC

Level-2A surface reflectance

Bands used:
B02, B03, B04, B08, B11, B12

Derived indices:
NDVI, NDBI, MNDWI, BSI

10 m analysis grid

SCL-based cloud/shadow masking

419,510 valid pixels



Sentinel-2C (False Color Composite)
B04 (Red) | B08 (NIR) | B02 (Blue)

0 5 10 km N

Source: Copernicus Data Space Ecosystem — Sentinel-2 documentation
https://documentation.dataspace.copernicus.eu/Data/SentinelMissions/Sentinel2.html

LANDSAT-9 · CURRENT PoC

Collection 2 Level-2 Surface Temperature (ST_B10)

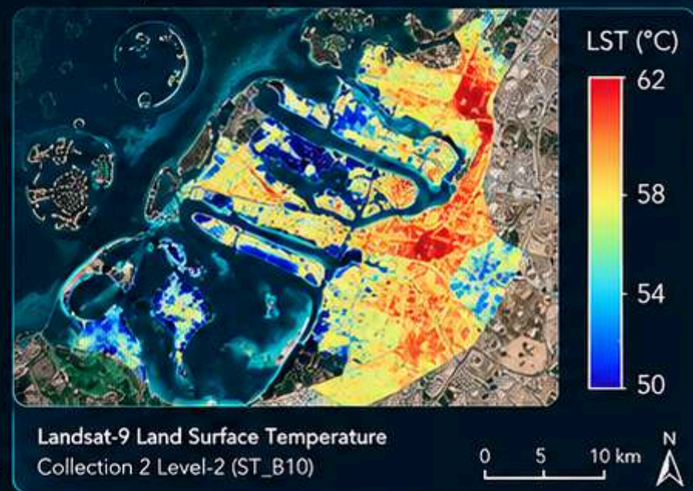
Collection 2 Level-2 Surface Temperature (ST_B10)

QA masking and product scale/offset applied

Converted to °C

46,616 valid LST cells

Observed LST is the model target, not ground-sensor truth.



Landsat-9 Land Surface Temperature
Collection 2 Level-2 (ST_B10)

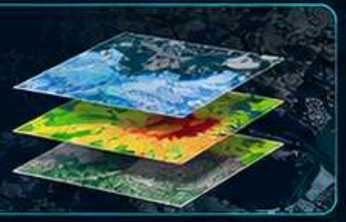
0 5 10 km N

Source: USGS Landsat Collection 2 Surface Temperature
https://www.usgs.gov/landsat-missions/landsat-collection-2-surface-temperature

CURRENT PoC

Sentinel-2C + Landsat-9 only

Used to derive spectral indicators, surface regimes, and 250 m planning-priority maps for the UAE study area.



FUTURE WORK — NOT USED IN CURRENT OUTPUTS

Explore hyperspectral Arab Satellite 813 and additional SAR/geospatial sources after data access and validation.



Study area: UAE. Team country: Jordan. Scene acquisition dates and product IDs should be added from the original notebook/source metadata.

HYPERSPECTRAL DATA

ARAB SATELLITE 813

High-resolution hyperspectral data for land and environmental monitoring.



Arab Satellite 813

- Hyperspectral imagery
- ~30 m resolution
- 150+ narrow bands
- Land & environment monitoring

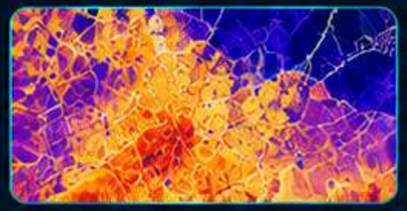
POTENTIAL BENEFITS FOR URBANTWIN AI

- Detailed vegetation analysis
- Surface material classification
- Additional environmental indicators
- Richer inputs for urban planning

EXAMPLE HYPERSPECTRAL LAYERS (CONCEPTUAL)



Vegetation Types



Vegetation Stress



Surface Materials



Soil Moisture

CURRENT STATUS

Not yet implemented in the current PoC.

07 · TECHNICAL WORKFLOW

FROM SATELLITE DATA TO PLANNING INSIGHTS

A geospatial AI pipeline that transforms Sentinel-2 and Landsat-9 data into spatially explicit urban indicators.

1 SATELLITE DATA

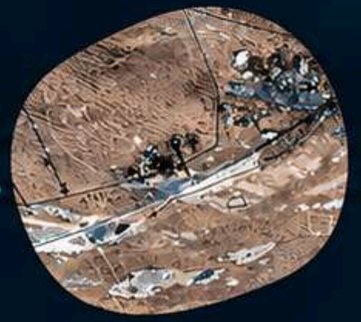
Two complementary EO products





2 PREPROCESSING

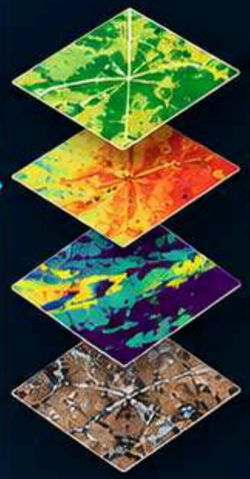
Quality-ready data for analysis



- Cloud/shadow masking
- Scaling & alignment
- Analysis-ready layers

3 FEATURE EXTRACTION

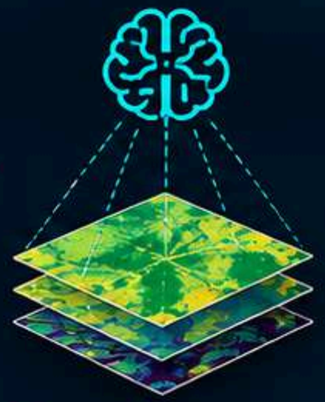
Derive key spatial indicators



- Vegetation index (NDVI)
- Land-surface temperature (LST)
- Surface clustering
- Additional spectral features

4 AI ANALYSIS

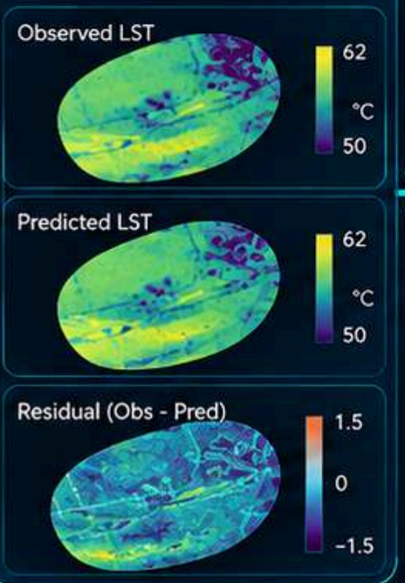
Identify patterns and predict LST



- Random Forest regression for LST prediction
- Unsupervised clustering for surface regimes

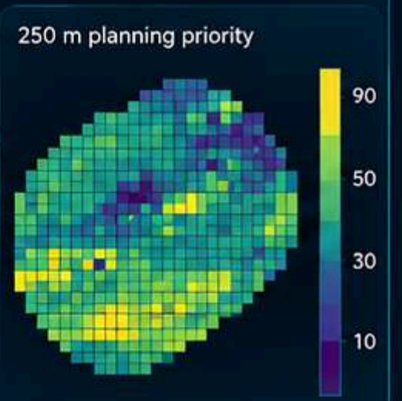
5 SPATIAL VALIDATION

Evaluate model performance



6 PLANNING OUTPUTS

Translate results into comparable planning cells



END-TO-END PIPELINE

Sentinel-2C + Landsat-9
EO inputs

→ **Preprocessing**
masking, scaling, alignment

→ **Features**
NDVI, LST, clustering, spectral bands

→ **AI Analysis**
clustering + LST prediction

→ **Validation**
spatial holdout evaluation

→ **Planning Layer**
250 m indicators

GOAL

Generate reliable, spatially explicit indicators to support urban planning and decision-making.

07 · TECHNICAL WORKFLOW

TWO EO PRODUCTS, ONE ANALYSIS CONTEXT

We combine Sentinel-2 optical surface information with Landsat-9 thermal observatiions to derive spatial indicators at 250 m for the UAE study area.



UAE STUDY AREA
Supplied AOIs, 3 km buffer
Total area: ~3,400 km²

01 SENTINEL-2C · OPTICAL SURFACE INFORMATION

Level-2A surface reflectance



- Level-2A product
- 10/20 m spatial resolution
- B02, B03, B04, B08, B11, B12 spectral bands

Provides spectral information for vegetation indices, surface features and clustering.

02 LANDSAT-9 · THERMAL REFERENCE

Collection 2 Level-2 Science Product

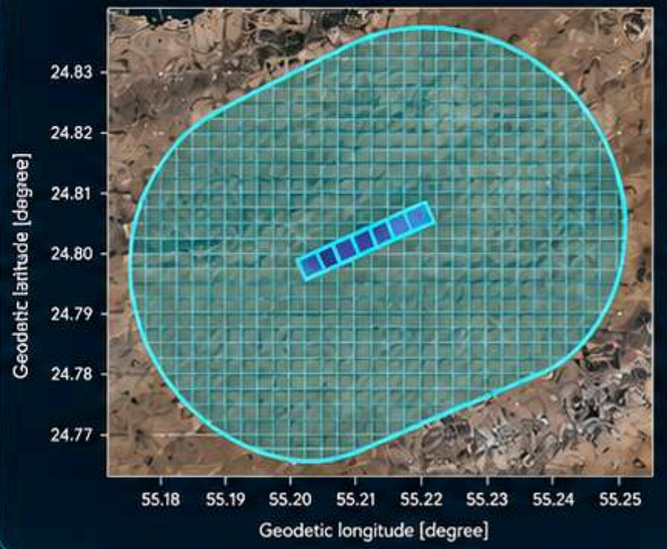


- Level-2 product
- 30 m spatial resolution
- ST_B10 LST product

Supplies the observed land-surface temperature target for model training and analysis.

03 STUDY CONTEXT AND ANALYSIS GRID

250 m planning grid and supplied AOI reference

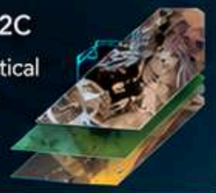


- Buffered study context (3 km)
- 250 m planning grid (612 cells)
- Supplied AOI segments (5)

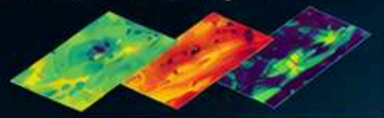
The study context, regular 250 m grid, and the original supplied AOI segments.

DATA TO ANALYSIS PIPELINE

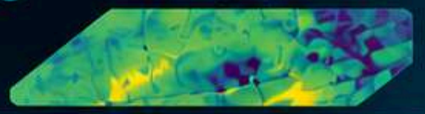
1. **Sentinel-2C**
   10/20 m optical reflectance



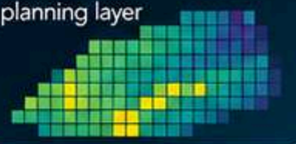
2. **Feature engineering**
   indices + surface regimes



3. **Landsat-9**
   30 m LST target grid



4. **Urban indicators**
   250 m planning layer



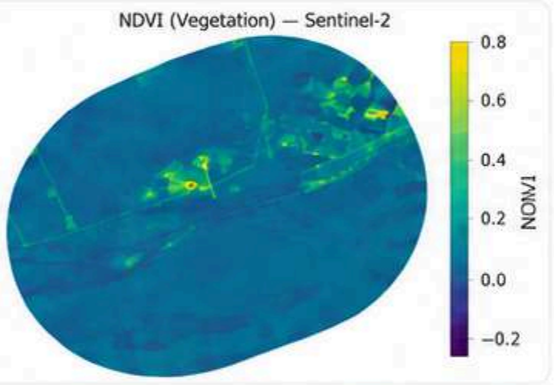
08 · ACTUAL POC OUTPUTS

SPATIAL INSIGHTS FROM THE CURRENT POC

Four key outputs generated using Sentinel-2C and Landsat-9 for the UAE study area.

01 · VEGETATION INDEX (NDVI)

Sentinel-2C



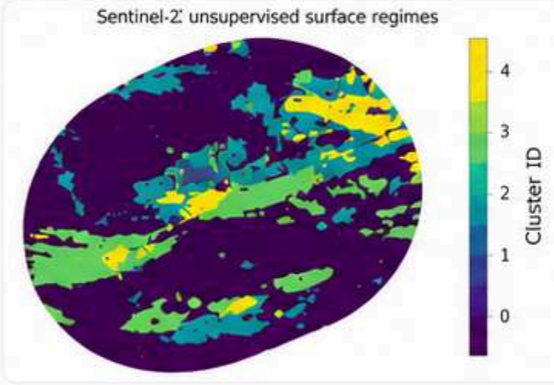
419,510
valid Sentinel-2 pixels

10 m
analysis grid

Highlights vegetation-sensitive spectral patterns across the study area.

02 · SURFACE-REGIME CLUSTERING

Sentinel-2C



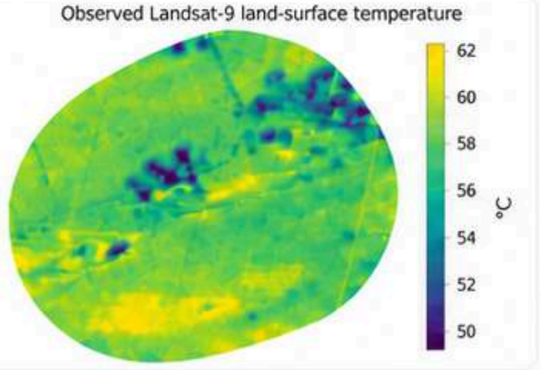
5
clusters

200k
fit pixels

Groups areas with similar spectral characteristics. Clusters are possible surface regimes, not validated land-cover classes.

03 · OBSERVED LAND-SURFACE TEMPERATURE (LST)

Landsat-9



48.53°C
minimum

58.81°C
median

62.33°C
maximum

46,616
valid LST cells

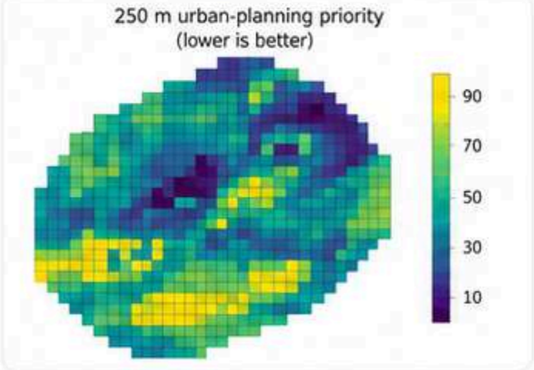
Processing in the notebook

- Use Landsat Level-2 ST_B10.
- Apply QA masking for invalid/cloud/saturated observations.
- Apply the scene metadata scale and offset.
- Convert the final temperature values to °C.

04 · 250 M PLANNING-PRIORITY MAP

Prototype planning score

```
Priority = 0.50 · Thermal Pressure + 0.30 · Vegetation Deficit + 0.20 · Heat-Associated Surface Fraction
```



Example high-priority cell - G004_012

96.58 priority score	**61.80°C** mean LST
99.82% thermal pressure	**83.36%** heat-associated surface fraction

Prototype priority score combining thermal pressure, vegetation deficit and heat-associated surface fraction. Not a validated heat-risk score.



Current PoC
UAE study area
Sentinel-2C + Landsat-9

KEY NUMBERS

419,510
valid Sentinel-2 pixels

10 m
Sentinel-2 analysis grid

46,616
valid LST cells (Landsat-9)

5
surface-regime clusters

200k
fit pixels

250 m
planning grid

KEY TAKEAWAY

UrbanTwin AI converts satellite observations into spatial indicators that can support further urban planning assessment.

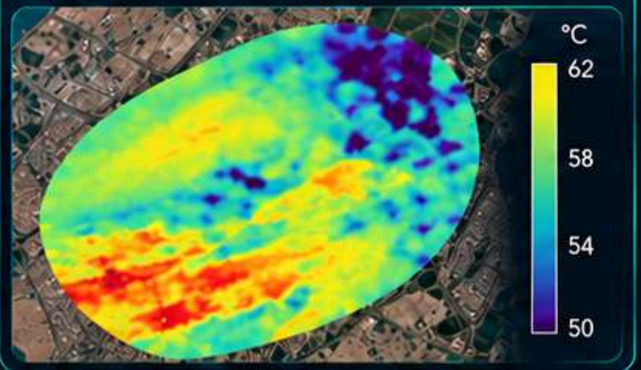
9 · MODEL VALIDATION

ACCURATE LST PREDICTION AT 250 M SCALE

We evaluate the model using observed vs. predicted LST and residual analysis to assess its reliability.

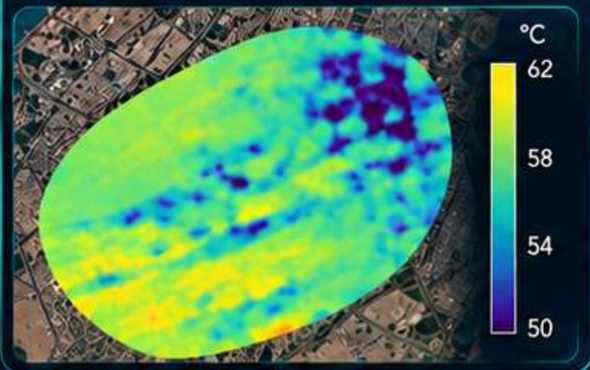
01 OBSERVED LST (LANDSAT-9)

Actual land-surface temperature



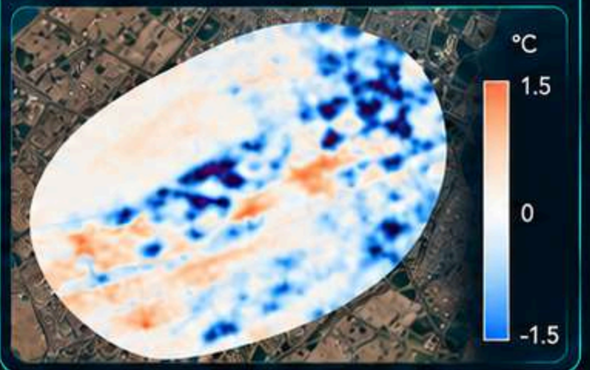
02 PREDICTED LST

Random Forest prediction (250 m)



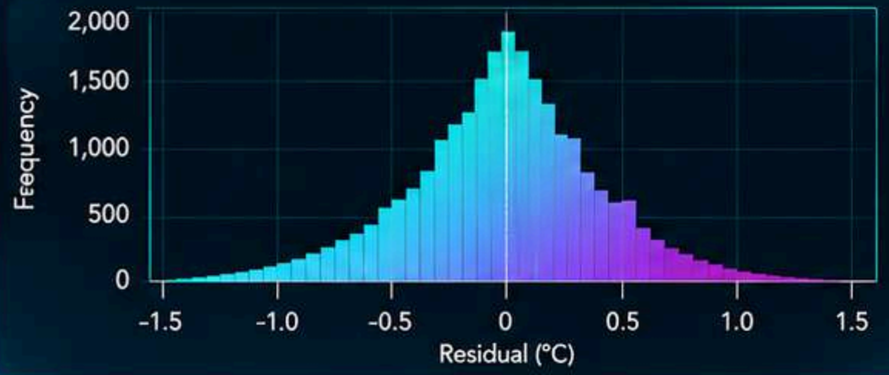
03 RESIDUALS (OBS – PRED)

Prediction error analysis



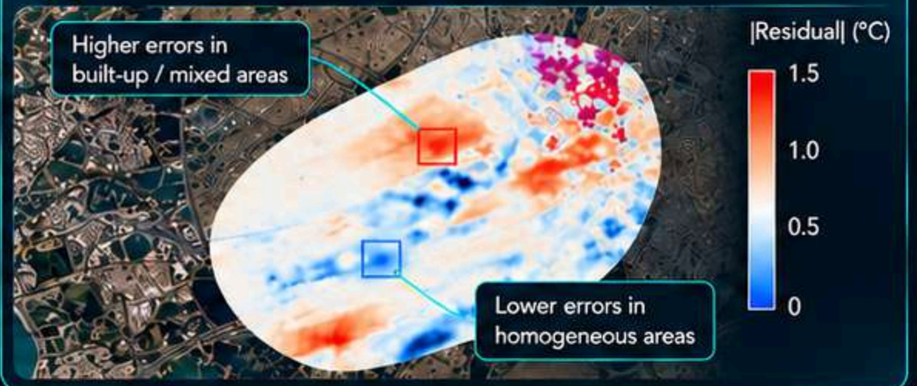
04 ERROR DISTRIBUTION

Distribution of prediction errors (residuals)



05 SPATIAL ERROR PATTERNS

Higher errors in certain surface regimes

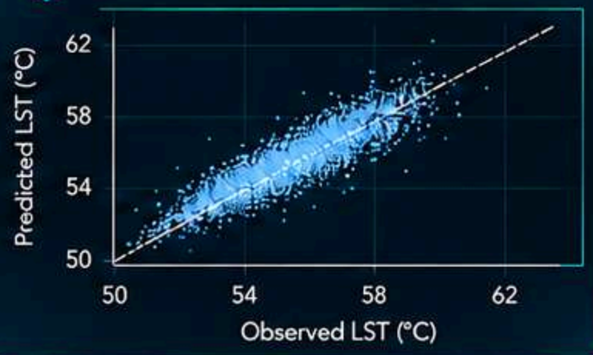


MODEL PERFORMANCE

0.512°C	0.713°C	0.532
MAE	**RMSE**	**R^2**
Mean Absolute Error	Root Mean Square Error	Coefficient of Determination

The model shows good performance in capturing spatial patterns of LST at 250 m resolution using Sentinel-2 and Landsat-9 data.

OBSERVED VS. PREDICTED LST



KEY TAKEAWAYS

- Low errors (MAE 0.512°C, RMSE 0.713°C).
- Moderate explanatory power (R^2 = 0.532).
- The model captures spatial trends and hotspots.
- Suitable for prototype urban planning analysis at 250 m resolution.

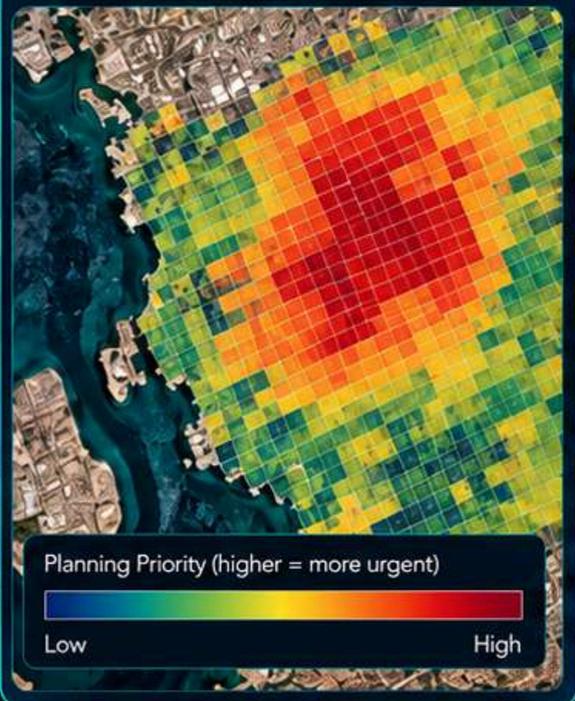
10 · PLANNING APPLICATIONS

FROM SATELLITE DATA TO ACTIONABLE URBAN PLANNING

UrbanTwin AI translates environmental insights into practical, location-based recommendations to support cooler, greener and more resilient cities.

01 PRIORITY AREAS

Identify where intervention is most needed (250 m grid).



02 TARGETED INTERVENTIONS

Different strategies based on local surface and vegetation conditions.



Vegetation & Shading

- Plant trees and green corridors
- Increase shaded areas
- Prioritize low-vegetation zones



Cooler Materials

- Use high-albedo materials
- Increase roof reflectivity
- Reduce heat absorption



Urban Design

- Improve ventilation corridors
- Increase green–gray balance
- Inform zoning and development planning

03 EXAMPLE PLANNING CELL

Detailed indicators for one high-priority 250 m cell.



Cell ID: G004_012

- Priority Score **96.58**
- Mean LST **61.80°C**
- Heat Pressure (0–1)
- Vegetation Deficit (0–1)
- Built-up Fraction (0–1)

POTENTIAL IMPACTS

Supporting more resilient and liveable urban environments.

Reduce Urban Heat

Lower surface and air temperatures in critical areas.

Increase Green Coverage

Support tree planting and green infrastructure.

Inform Sustainable Urban Design

Guide zoning, materials and open-space planning.

Support Evidence-Based Decision Making

Provide spatial insights for policymakers and planners.

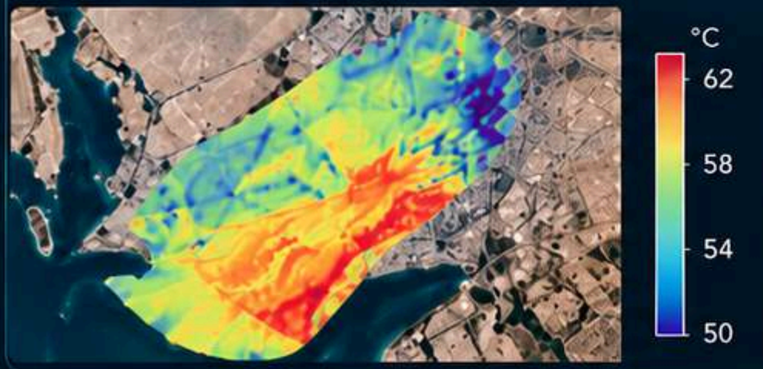
NEXT STEPS

- Refine and validate the planning priority framework with additional data.
- Explore higher-resolution and real-time data sources (e.g. Arab Satellite 813).
- Integrate results into urban planning workflows and decision-support tools.
- Engage with local stakeholders for pilot implementation in selected areas.

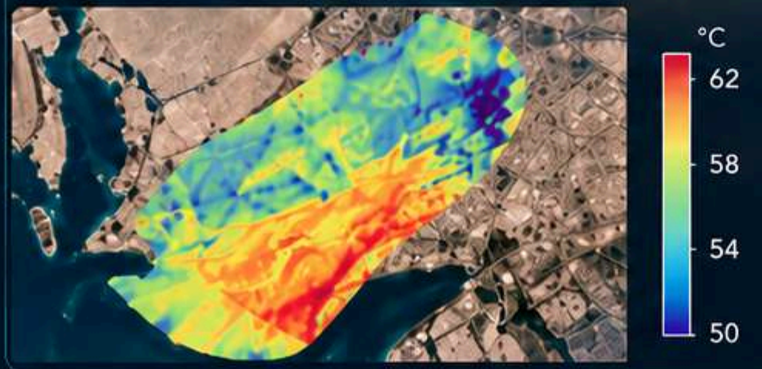
11 • MODEL VALIDATION & LIMITATIONS

MODEL VALIDATION & LIMITATIONS

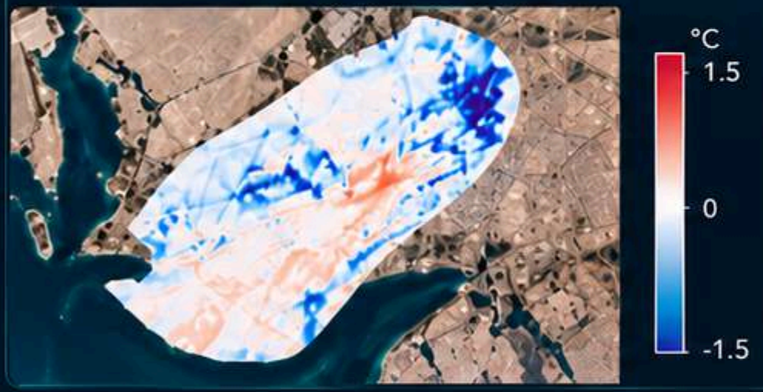
01 Observed LST (Landsat-9)



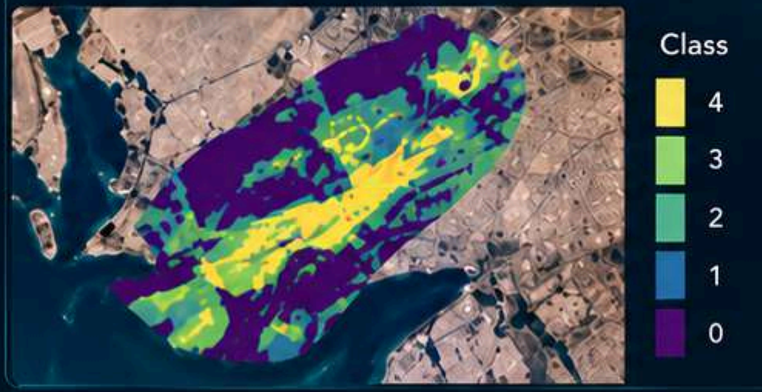
02 Predicted LST (Random Forest)



03 Residuals (Observed – Predicted)



04 Surface Regime Clusters (5 classes)



MODEL PERFORMANCE

0.512°C	0.713°C	0.532
MAE	RMSE	R^2

VALIDATION SETUP

300 m
spatial blocks

34,899	training samples
11,670	test samples
46,569	total samples

KEY LIMITATIONS

- Validated against Landsat-derived LST only
- Moderate model performance
- Single observation period
- LST is not air temperature
- Surface regimes are unsupervised
- Priority score is a heuristic indicator
- Results not yet validated for Jordanian cities

12 • INCUBATION ROADMAP

FROM POC TO REAL-WORLD IMPACT

A clear path to validate, develop, and scale UrbanTwin AI into a practical decision-support tool for more resilient and sustainable cities.

01 REFINE & VALIDATE
Oct – Dec 2026



- Improve model and features
- Add more satellite data and dates
- Test in additional city contexts (including Jordan)

02 DEVELOP MVP
Jan – Feb 2027



- Build MVP (**February 2027**)
- Interactive web-based prototype
- Visualize heat hotspots and planning indicators
- Integrate user feedback

03 PILOT & PARTNERSHIPS
Mar – Jun 2027



- Run pilot studies with local partners (e.g. municipalities, universities)
- Validate usefulness for real planning decisions
- Gather feedback and refine

04 SCALING & IMPACT
Jul 2027 onwards



- Expand to more cities and regions
- Explore integration with ManzelTwin
- Enable long-term monitoring and scenario analysis
- Support data-driven, climate-resilient urban planning

EXPECTED OUTCOMES

- Validated model
- Functional MVP (Feb 2027)
- Use cases with local partners
- Scalable solution for MENA region

LONG-TERM VISION

A regional urban intelligence platform that combines satellite data, AI, and spatial analytics to support cooler, greener and more resilient cities.

NEXT STEPS FOR INCUBATION

FROM INSIGHTS TO A DIGITAL TWIN

Building on the PoC, UrbanTwin AI aims to evolve into a Digital Twin–enabled planning platform, turning satellite-derived insights into immersive and actionable urban intelligence.

01 EARTH OBSERVATION

Sentinel-2 and Landsat-9 data



02 URBANTWIN AI

Surface indicators, thermal prediction and planning-priority outputs



03 FUTURE MANZELTWIN INTEGRATION

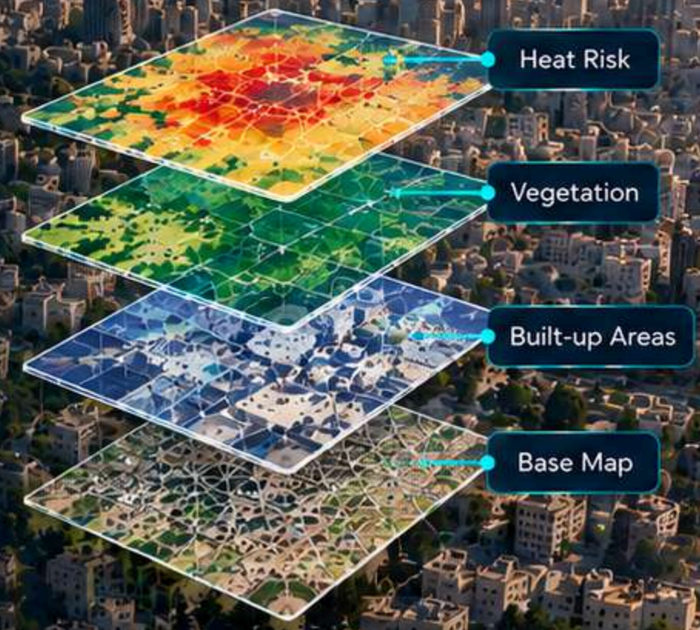
Visualize validated geospatial layers in a virtual urban environment



04 PLANNING DECISION SUPPORT

Explore locations, compare indicators and support planning assessments





CURRENT POC

- Satellite data analysis
- Thermal prediction
- Planning-priority maps

INCUBATION PHASE
(3–6 Months)

- Refine models and validation
- Enhance spatial layers
- Prototype Digital Twin visualization

LONG-TERM VISION
(6+ Months)

- Full Digital Twin integration (ManzelTwin)
- Scenario exploration and what-if analysis
- Collaboration with planning stakeholders